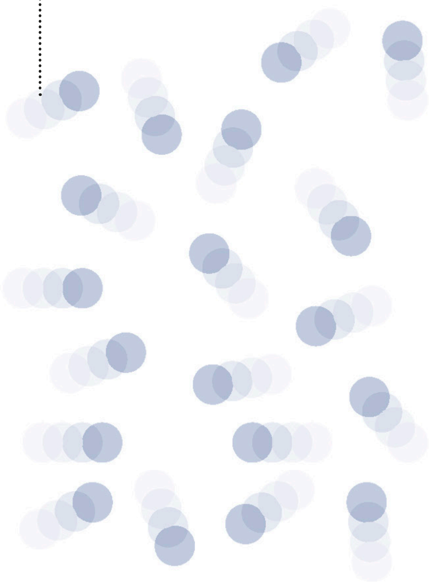
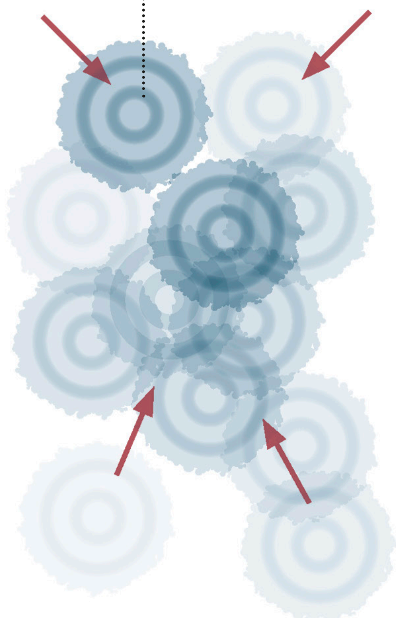
BOSONIC GAS

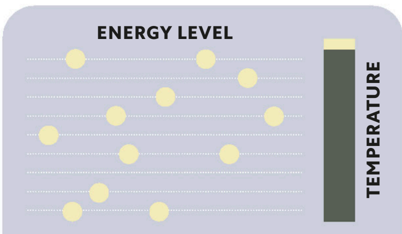
Because of the way the quantum spin property adds up (see p.68), atoms containing the right number of fermion particles (spin ½) can behave as bosons with whole-number spins.

OVERLAPPING WAVES

As a sparse gas of bosonic atoms is cooled to very low temperatures, their wave functions expand and begin to overlap with each other.

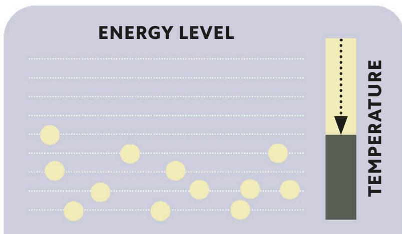






Normal temperatures

At high temperatures, the boson gas seems outwardly normal, with particles at a wide range of energy levels—although some levels are shared.



Extreme cooling

As temperatures fall and the particles lose kinetic energy, their range of possible energy states is reduced. More particles start to share the same state.